



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to MODERN AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

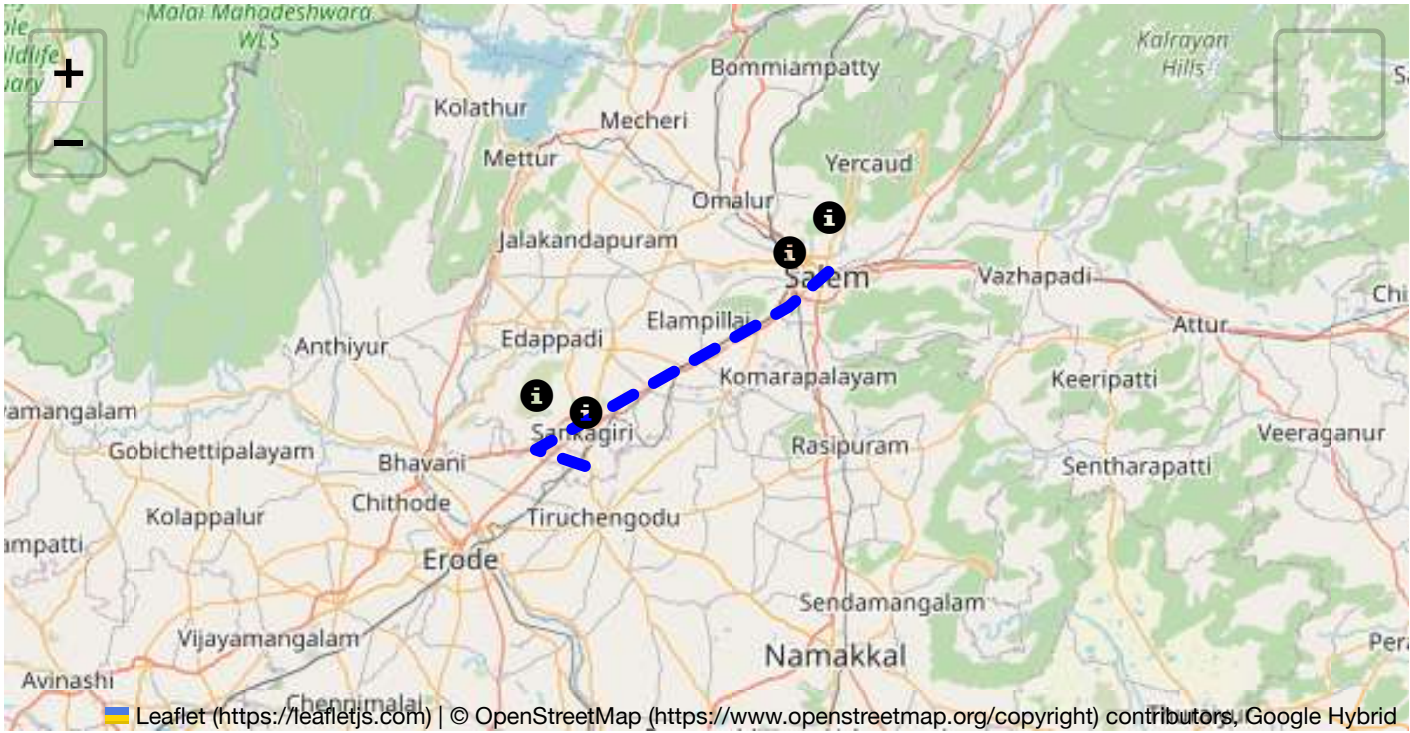
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

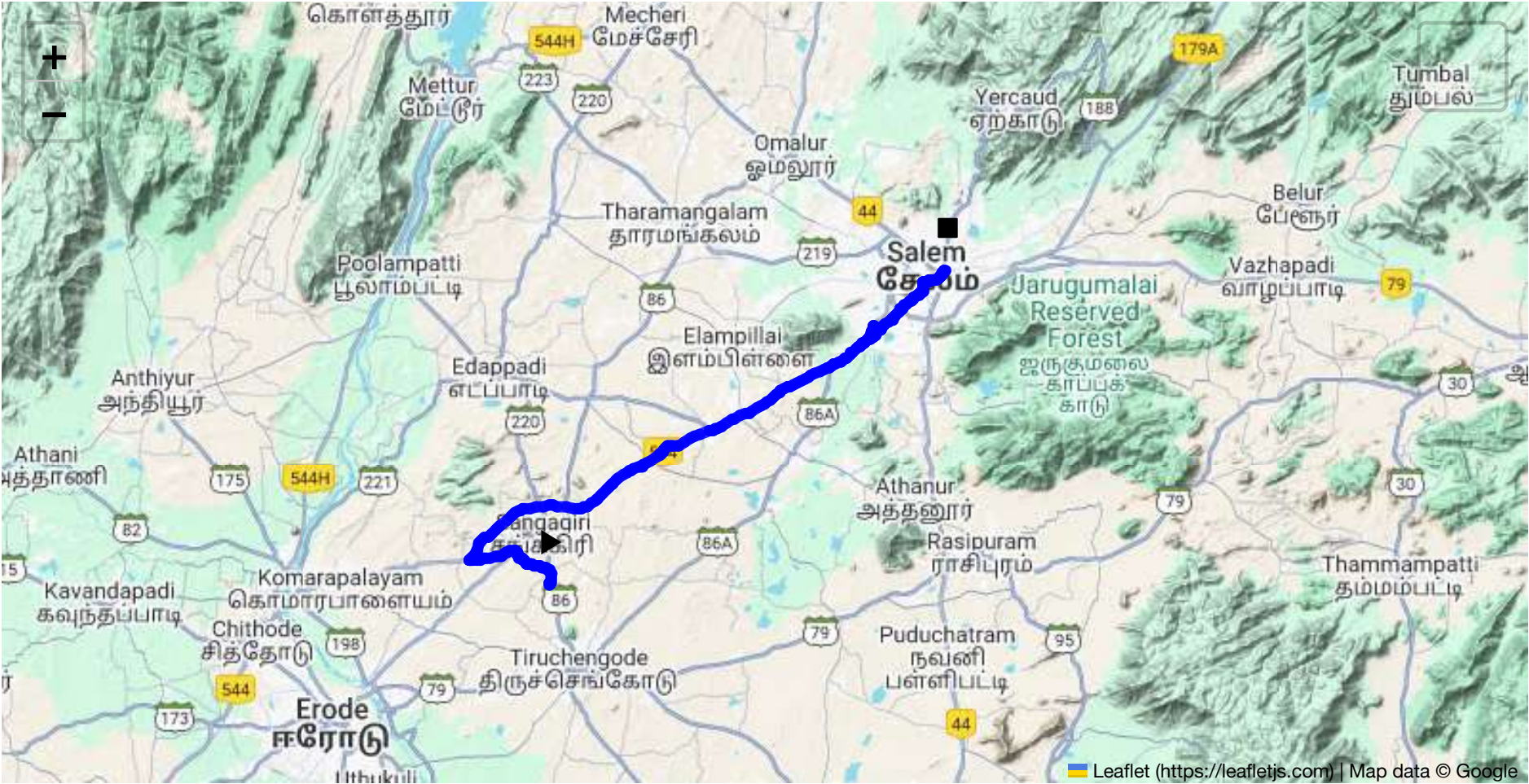
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 56.97 km
Estimated Duration: 1.3 hours
Adjusted Duration (Heavy Vehicle): 1.6 hours

Start: (11.4381, 77.8734)
End: (11.659975, 78.159036)



Welcome to the Journey Risk Management Study

Overview of the Route Map

The route spans approximately 56.97 kilometers, primarily traveling through rural and semi-urban areas. Starting from Sangagiri, Tamil Nadu, it progresses through towns including Padaiveedu and Kondalampatti before reaching Salem. The route involves a mix of National and State Highways, with transitions through smaller town roads.

Typical Weather Conditions and Potential Weather-related Hazards

- Weather Conditions:** The area generally experiences a tropical climate. During the monsoon months (June to October), heavy rains can lead to slippery roads and reduced visibility.
- Hazards:** Flooding in low-lying areas could occur during intense rains, potentially causing road closures. In high summer (April-June), heat can lead to road surface deterioration.

Analysis of Traffic Patterns

- Peak Hours:** Traffic tends to peak during mornings (8-10 AM) and evenings (5-7 PM), particularly near marketplaces and entry/exit points of towns.
- Congestion-Prone Areas:** Critical areas include town centers like Kondalampatti, where roads might be narrow, and intersections near the Salem bypass.

Assessment of Road Quality and Infrastructure

- Road Quality:** Varies from well-maintained highways to less maintained town roads, where potholes and uneven surfaces are prevalent.

- **Infrastructure:** Adequate signage is in place on highways but may be less so in towns. Lighting is sufficient on main roads but can be sparse on local roads.

Suggestions for Alternative Routes for Emergencies

- Consider using the NH44 as an alternative for bypassing congested areas or during adverse weather conditions, though it may increase travel time slightly.
- Internal roads through nearby villages may be used for certain stretches to avoid heavy congestion.

Summary of Local Regulations Affecting Hazardous Material Transport

- Mandatory permits and appropriate labeling are required for transporting hazardous materials.
- Restrictions on the movement of such vehicles through highly populated areas during peak hours should be observed.

Overview of Historical Incidents

- **Historical Incidents:** There have been occasional reports of accidents involving heavy vehicles, mostly due to driver fatigue and road conditions.
- **Hazardous Materials:** No major spillage incidents reported recently, but precautionary measures should always be adhered to.

Environmental Considerations and Sensitive Areas

- **Sensitive Areas:** Agricultural land near the routes that might be impacted by spills.
- **Considerations:** Care should be taken near water bodies and cultivation areas to prevent contamination.

Analysis of Communication Coverage

- **Coverage:** Generally good along major highways, but potential dead zones exist as the route traverses through rural areas and between towns.
- **Recommendation:** Equip drivers with alternative communication devices such as satellite phones for emergencies.

Estimated Emergency Response Times

- **Rural Segments:** Response times can be delayed, averaging 30-45 minutes due to distance from emergency services.
- **Urban Areas:** Quicker, approximately 15-20 minutes due to closer proximity to emergency facilities like hospitals and fire stations.

Overall Summary of Risk Assessment

This route presents moderate risks due to variable road conditions and traffic patterns. Key hazards include congestion in towns, weather impacts, and variable road quality. Proper preparation, such as adherence to regulations, maintaining communication, and alternative route planning, can mitigate these risks. Continuous monitoring of weather conditions and real-time traffic updates is essential for safe passage.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Roundabout	High	11.65514, 78.15419	15 KM/Hr
2	Roundabout	High	11.65653, 78.15533	15 KM/Hr
3	U-Turn	High	11.6193714, 78.1124012	10 KM/Hr
4	U-Turn	High	11.618013, 78.1109532	10 KM/Hr
5	Turn	Medium	11.43822, 77.87348	30 KM/Hr
6	Turn	High	11.43968, 77.87345	15 KM/Hr
7	Turn	High	11.44029, 77.87544	15 KM/Hr
8	Turn	Medium	11.45358, 77.85781	30 KM/Hr
9	Turn	Medium	11.45353, 77.85697	30 KM/Hr
10	Turn	Medium	11.45839, 77.85143	30 KM/Hr
11	Turn	Medium	11.46053, 77.85034	30 KM/Hr
12	Turn	Medium	11.46307, 77.84941	30 KM/Hr
13	Turn	Medium	11.46316, 77.84921	30 KM/Hr
14	Turn	High	11.45542, 77.81725	15 KM/Hr
15	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
16	Blind Spot	Blind Spot	11.62145, 78.10740	10 KM/Hr
17	Turn	High	11.62130, 78.10776	15 KM/Hr
18	Blind Spot	Blind Spot	11.61217, 78.10439	10 KM/Hr
19	Blind Spot	Blind Spot	11.61937, 78.11240	10 KM/Hr
20	Turn	Medium	11.61924, 78.11237	30 KM/Hr
21	Turn	Medium	11.61812, 78.11108	30 KM/Hr
22	Blind Spot	Blind Spot	11.61807, 78.11090	10 KM/Hr
23	Turn	Medium	11.64844, 78.14338	30 KM/Hr
24	Turn	High	11.65197, 78.14241	15 KM/Hr
25	Turn	Medium	11.65752, 78.15667	30 KM/Hr
26	Turn	High	11.65998, 78.15751	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
14	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
16	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
17	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
18	clinic	Salem Care Foundation	11.6371946, 78.1313713	30 km/h	Medium
19	hospital	Aravind Eye Hospital	11.641927, 78.1392347	30 km/h	Medium
20	police	Annadanapatti B 4 Police Station	11.6424031, 78.1404779	30 km/h	Medium
21	clinic	Shri Bharthi Nursing Home	11.6420069, 78.1424099	30 km/h	Medium
22	hospital	Ayyappa Hospital	11.6524111, 78.1456836	30 km/h	Medium
23	hospital	Pramilagurunathan Hospital	11.6544063, 78.1464484	30 km/h	Medium
24	hospital	Salem Government Hospital	11.6554386, 78.1520712	30 km/h	Medium
26	police	Shevapet B 2 Police Station	11.6514754, 78.1522608	30 km/h	Medium
27	hospital	Corporation Maternity Hospital	11.6524437, 78.154675	30 km/h	Medium
31	hospital	Shri Suraksha Speciality Medical Centre	11.6590215, 78.1587125	30 km/h	Medium
32	hospital	Sri Sellappa Hospital	11.65978, 78.154787	30 km/h	Medium
33	hospital	Kerrison Memorial Hospital	11.6604257, 78.1579573	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
34	clinic	Salem Heart Clinic	11.660386, 78.1556973	30 km/h	Medium
36	hospital	A.G.P Hospital	11.6608027, 78.1554709	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
15	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
25	school	CSI Hr. Sec. School	11.6536296, 78.1529373	30 km/h	Medium
28	school	Govt Boys Hr. Sec. School	11.6550838, 78.155882	30 km/h	Medium
29	school	Govt Girls Hr Sec School, Fort	11.6547568, 78.1578092	30 km/h	Medium
30	school	Govt Girls Hr Sec School	11.6550993, 78.1575234	30 km/h	Medium
35	school	Corporation High School	11.6608771, 78.1573617	30 km/h	Medium
37	college	GOVT. ARTS COLLEGE	11.6640046, 78.1596148	30 km/h	Medium
38	school	Bala Bharathi Matric Hr Sec School	11.6619666, 78.1624808	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.65514, 78.15419



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.65653, 78.15533



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 11.6193714, 78.1124012



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 11.618013, 78.1109532



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45358, 77.85781



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45839, 77.85143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46053, 77.85034



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46316, 77.84921



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62145, 78.10740



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.62130, 78.10776



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61217, 78.10439



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61937, 78.11240



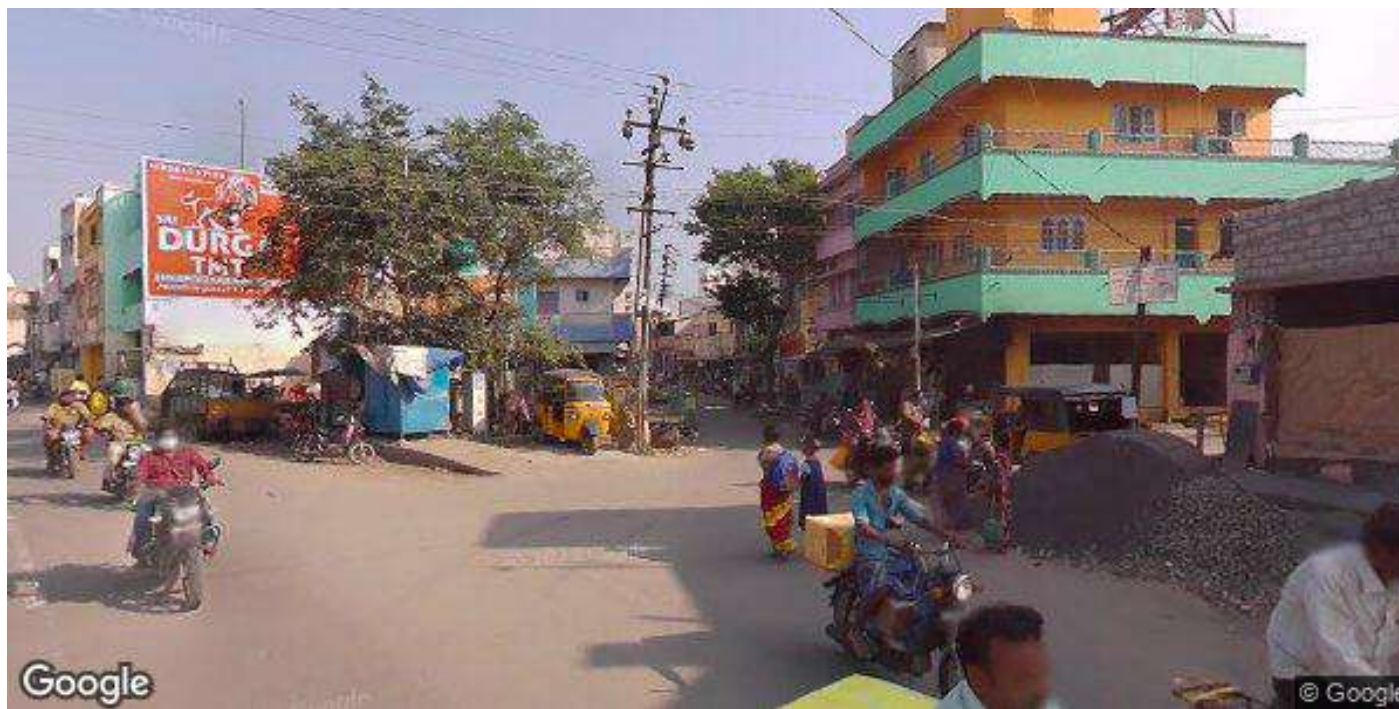
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61924, 78.11237



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61812, 78.11108



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61807, 78.11090



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64844, 78.14338



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65197, 78.14241



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65752, 78.15667



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65998, 78.15751
